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OFFER
No. 801926/REV.3
FOR
FORGING MACHINE SKK-10
(EXPANDABLE FOR TURNKEY CELL)

Type: GFM SKK-10

Dossier: Project #1P02296.20

Layout: GFM #1P02296.20.02

Customer: HYUNDAI-WIA Corp.

Representative: Mr. MK. Kim, Jin-A Commerce & Co.

Location: HWI Chennai, India

Your contact: Franz Hofer, GFM-Sales
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Date: 18.10.2021
(Rev.1: 16.12.2021, Rev.2: 31.12.2021, Rev.3: 11.01.2022)

Our reference: VK-Hofer/zec

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1. PROJECT DESCRIPTION

1.1. Method of operation

The forging machine according to enclosed layout # 1P02296.20.02 is suitable for cold forging of hollow drive shafts (MTS).

The blanks are taken into the blank container manually. At the end of the blank container the blanks are separated automatically and the inner surface will be sprayed with an air/oil mist.

From there the robot takes the blank to the manipulator of the forging machine SKK-10. The machine is laid out for full automatic radial forging, in 'feed mode' as well as 'plunge-in mode'.

The robot is equipped with a 'double gripper system' and thus will take the forged shaft out of the forging center axis, while simultaneously loading the next blank to the manipulator of the machine.

While forging the next part, the robot takes the forged MTS component to the finished parts box (unloading box)

The CNC-control allows full automatic forging with process control and listing of pre-adjustable parameters.

The forging machine works fully automatically, beginning from the handling of the blanks to the forging machine and consequently performing the forging via preselected program, then handling the forged shafts with the robot to the unloading box.

The robot can also take randomly a part and place it into an optional 'Inspection part discharge belt' for QC of the work piece geometry while the machine keeps running in automatic mode.

1.2. GFM-Radial forging process

The direct drive motor (Torque motor) in the manipulator prevents, any undesired torsion torque acting upon the work piece, during the hammer contact.

The technology of GFM radial forging enables to accomplish long forging-contours, as well as production of complex and precise inner-contours.

During forging the work piece is cooled with circulating water, therefore contamination of the workpiece with oil emulsion, as well as general oil pollution is prevented.

1.3. Sample production and forging trials

GFM can produce sample parts to customer specification for development purposes and process validation, prior to machine delivery. Two demo-machines are permanently available for manufacturing prototypes and small initial series orders.

2. TECHNICAL DATA OF FORGING MACHINE SKK-10

2.1. MECHANIC

Number of forging hammers	4
Max. forging force	1250 kN
Max. axial force	40 kN
Number of forging strokes	appr. 1200/min.
Max. blank diameter	65 mm
Max. work piece length	850 mm
Min. work piece length	350 mm
Max. work piece weight	6 kg
Infinitely variable feed speed of chuck head	1-500 mm/sec.
Infinitely variable feed speed of hammers	0,02-8 mm/sec.
Adjustable range of the forging ram in diameter	60 mm
Total weight of machine	appr. 30 t
(without special equipment and tools)	

Technical data are subject to change!

2.2. ELECTRIC

- 1 Main drive motor, 90 kW (incl. soft start device)
- 1 Motor for work piece rotation on chuck head, ~8kW (torque motor)
Various motors for lubrication, approx. 15 kW
Various motors for hydraulic, approx.. 22 kW
- 1 Combined control cabinet with separate sections for power unit
(Electric supply, motor starters, soft starter) and CNC with
integrated programmable logic (PLC)
Both, the CNC supply (230 VAC) as well as the control voltage
(24 VDC) are equipped with non-interruptible power (UPS)
- 1 Control desk, complete with integrated industrial Box-PC &
19" TFT-monitor for machine operation, programming and display
of various production data, such as: feed rates, pressures, forging
power, forging force, multi-channel oscilloscope, fault messages etc.
Machine operation and program handling is performed
on a 'Windows 10®' operating system.

Operating voltage 3 x 415 VAC, 50 Hz

Control voltage

- for motor starters and cabinet illumination..... 230 VAC, 50 Hz

- for solenoids, limit switches and push buttons 24 VDC

2.3. CNC-CONTROL

Consists of PC with integrated free programmable logical control (PLI) and axis controller.

The CNC 6000 GP is an especially developed and optimized system for the use on GFM-Radial forging machines with NC axis:

Axis A = manipulator (chuck head)

Axis B = counter holder

Axis L = hammer adjustment, rams 1-3

Axis R = hammer adjustment, rams 2-4

Axis D = forging mandrel, chuck head side

Operating and programming is accomplished with interactive communication on the monitor.

Forging programs and setting data will be stored in a hard-disk memory. Program input can be done from the CNC-keyboard or by using the DNC-interface (ISO/R 840-1968 formatted protocol).

Input accuracy:

0.1 mm for axis A

0.1 mm for axis B

0.01 mm for axis L & R

0.1 mm for axis D

Positioning accuracy:

for manipulator axis (A) $\pm 0,2$ mm

for counter holder axis (B) $\pm 0,2$ mm

for hammer axis (L & R) $\pm 0,02$ mm

for mandrel axis (D) $\pm 0,2$ mm

2.4. CONNECTION DATA

1. Electric

motors and devices 180 kVA, 415 VAC $\pm 5\%$
3 phase 50 Hz $\pm 2\%$

2. Cooling Water – closed circuit

outflow of forging machine max. 5 m³/h
(at a temperature difference of appr. 10°C)
inlet pressure 5 - 6 bar
differential pressure (inlet/outlet) approx. 2 bar
chlorides, sulfates, nitrates max. 50 mg/l
carbonate hardness max. 25 dH
pollution..... max. 100 mg/l
dirt grain size max. 0.3 mm

3. Compressed Air

connecting point
consumed air approx. 0,5 N m³/min.
supply pressure 5-6 bar

The air must be free of aggressive gases.

An equipment for separating of water and solid substances has to be provided by customer.

4. Consumption Data

(based on 2-shift functions)

lubricating oil appr. 600 l/2years
hydraulic oil appr. 250 l/3years
guide way oil 40 l/year
grease..... 15 kg/year

GFM recommends yearly inspection of hydraulic- and lubrication oil.

2.5 SOFTWARE FEATURES (GFM CNC 6000 GP)

The machine operation is 'Windows 10®' based and takes place exclusively via monitor and includes the following:

- operating of machine
- processing of forging programs (stored on hard disc)
- input of machine parameter (stored on hard disc)
- display of actual value
- display of nominal value
- display of fault messages and info for production data
- production run status
- tool management
- automatic machine start procedure
- oscilloscope function for datalog and service, as well as simultaneous acquisition of measuring values.
- trouble shooting menu with priority task function
- display of actual machine performance values for hydraulic and electric parameters
- automatically diameter compensation in reference to forging box temperature.
- automatically diameter compensation in reference to forging force
(this ensures tighter tolerances particularly with varying yield strength of blank material)

3. TECHNICAL DESCRIPTION FOR FORGING MACHINE SKK-10

3.1. FORGING BOX WITH DRIVE UNIT AND OTHER EQUIPMENT

The forging box consists of two rigid housing parts.

These housing parts are flanged together in vertical plane. In the forging box, 4 ram units are arranged axial-symmetrically, on which the forming tools are fixed.

The rams are actuated by 4 synchronously driven eccentric shafts.

The drive system for the eccentric shafts is integrated in the forging box.

The drive of the ram units is done via a 3-phase AC motor and a pinion.

The drive gear which are placed on the eccentric shafts are synchronously driven by this pinion.

The mechanically driven work piece adjustment is integrated in each ram unit.

The NC-controlled tool adjustment units are driven by 2 hydraulic motors which are synchronized and therefore a common adjustment of all 4 forging tools is performed.

Manipulator and manipulator bed (A)

The manipulator head slides on guide ways of the manipulator bed.

It clamps the work piece and provides the rotation and feed movement according the pre-selected forging program.

Feed force, feed movement as well as rapid traverse are given by a hydraulic cylinder.

The rotation is provided by an electrically driven direct drive motor (Torque motor), which is a robust-, multipole-, water cooled synchronous motor, with a fixed-, non-rotating coil and a permanent magnetic rotor.

The forging box, manipulator and counter holder are mounted on a common frame which ensures a high axis accuracy between manipulator and tools.

Counter holder (B)

On the gear box side of the machine a counter holder is mounted.

The work piece is clamped between the counter holder and the chuck head.

The value of the axial chucking force can be adjusted in order to adapt to the various forging conditions during radial forging.

Mandrel (D)

On the rear of the chuck head a forging mandrel is mounted (mandrel D). The forging mandrel is connected with a hydraulic cylinder via mandrel rod. The mandrel position can axially be adjusted according to program requirements.

Mandrel (D_B)

On the rear of the counterholder a forging mandrel is mounted (mandrel D_B). The forging mandrel is connected with a hydraulic cylinder via mandrel rod. The mandrel position can axially be adjusted according to program requirements.

Hydraulics

The hydraulic station consists of the hydraulic oil tank, the pumps and control elements. The hydraulic oil is held at constant temperature by an oil water heat exchanger.

Lubrication

The lube oil station consists of the lube oil tank, the pumps, filters, control elements as well as oil water heat exchanger which keeps the lube oil at constant temperature.

Control of the machine

The control of the machine, as adjustment for the forging hammers, selection of chuck head rapid traverse and feed motions, movement of forging mandrel, etc. is performed electro hydraulically, via a CNC-control system.

Safety Devices

The machine is protected against exceeding the maximum admissible forging power and/or forging force by means of an electronic overload safety device.

When exceeding the max. admissible forging power or forging force, the forging tools will open automatically, the mandrel withdraws, and subsequently all hydraulic movements come to a stop.

Workpiece cooling

The forging hammers and the workpiece are surrounded by coolants (cooling water with anti-corrosive additive) during the forging procedure, which eliminates unintended heating of the workpiece during forging.

This ensures tight tolerances, as well as best repeatability on workpiece tolerances.

4. TECHNICAL DESCRIPTION OF ADITTIONAL EQUIPMENT & TOOLS

4.1. Work piece handling device for fully automatic operation sequence,

consisting of:

- 6-axis robot, FANUC M-20iA (20kg, 1811mm), incl. 'R-30iB-AR Control, Double-parallel gripper 'Schunk ©' with pneumatic fingers, increased wet protection and pedestal, as well as a Profibus slave board.
- 2-side light grid protection system around unloading box area, to ensure a safe exchange handling when box is finished.

4.2. Noise protection cabinet - (optional: safety fence)

covering SKK machine, hydraulic station and robot, complete, for the noise decrease. This will be localized by HWI, based on GFM layout design.

Full illumination-, intake- and exhaust fan, 1 sliding door– and 4 swivel doors with el. safety locks. (Electrical safety locks & E-stop wiring will be supplied by GFM)

4.3. Blank container, compl. incl. separating device and automatic spray unit

consisting of:

- Firm frame design, with level adjustments
- Bunker magazine for blanks, capacity: > 100 parts @ Ø35mm
- Take over directly from sensor controlled lifting rail (w/o rolling down)
- Length adjustment with spindle, symmetrical on both sides
- Hardened steel sheet metal on bunker sidewalls for high wear resistance
- Axial alignment with pneumatically operated cylinder
- Custom designed spray nozzle for even- and sparely spray distribution

4.4. Work piece cooling system

- The forging hammers and the work piece are surrounded by coolants (cooling water with anti-corrosive additive) during the forging procedure, which eliminates unintended heating of the work piece during forging.
- A coolant pump serves for circulation of the cooling water, which is supplied from a separate tank. The cooling water passes through an oil skimmer and is kept at constant temperature by heat exchanger. An automatic filtering station ensures proper cleaning of the water, eliminating the abrasion particles and during hammering.

4.5. Pre-Spline diameter inspection: (OPTION, only required at expansion)

After one side of shaft is forged, the robot takes it to a 'Cleaning & Pre-spline diameter QC-station' on the SKK-10 forging machine. If the pre-spline dia is out of tolerance, the robot will automatically discharges the MTS and this evidence will show up on the MMI screen, too.

4.6. Tooling:**4.6.1 General forging tools for all shaft applications:**

- 1 set (4 pcs.) base plates, including tension bolts
- 1 set 4 pcs.) hammer clamps, including bolts and bushings
- 2+2 sets (total 4 pcs.) robot gripper (Schunk © parallel grippers)

4.6.2 Forging tools for one standard MTS shaft (in 2 sequence operations)

- 1 set (4 pcs.) forging hammers (for feed forging OP1 or OP2)
- 2 pairs (4 pcs.) of chuck jaws
- 1+1 (2 pcs.) forging mandrels
- 2 counterholder inserts
- 1 set of auxiliary tools (stop, adapter, mandrel rod etc.)

4.6.3 Tools for machine service and maintenance (N/A, OPTION only)

- 1 set of special tools for SKK-10 alignment and inspection
- 1 set of various commercial wrenches & sockets
- 1 set of various gauges and monitoring devices

5. Scope of supply and Cost Summary (for Single SKK-10, expandable)

Pos.	Item description	Further info	Cost
01	1 no. Blank container, incl. separating device and 1 no. Automatic ID spray unit, 1 no. Dia inspection unit (prepared) & 2-side light grid protection system around unloading box area	Sound enclosure not included, except door safety locks & E-stop wiring supplied by GFM.	EUR 71.620,00
02	1 no. GFM machine, Model SKK-10 for Radial cold forging MTS. (prepared for expansion to turnkey cell)	New generation, with torque drive system.	EUR 1,511.880,00
03	1 no. of robot, FANUC M-20iA/1,81, Control R-30iB-AR (incl. double gripper 'Schunk' & consideration for later expansion)	Including Schunk [®] parallel grippers, programming and machine interfacing	EUR 128.760,00
04	Forging tooling, complete for 1 no. MTS, 2 operations, incl. engineering & development. (w/o machine service tooling)	Including 1 set general forging tools for both SKK-10 machines	EUR 30.910,00
05	Trial forging of prototypes at GFM, Pre acceptance and customer training, incl. MFU-evaluation and QC	Sufficient blank tubes must be provided free of cost by Hyundai-Wia	EUR 12.770,00
06	Supervision of assembly, start-up and Final acceptance test at Hyundai-Wia, Chennai plant	For one SKK-10, incl. HMI for FANUC robot, blank container & workpiece cooling system	EUR 95.600,00
07	Dismantling, seaworthy packing, transportation - FOB European seaport.		EUR 42.650,00
08	1 no. HYFRA chiller, Sigma 160 (prepared for expansion up to 2 GFM machines, SKK-10)	Incl. piping and electr. wiring	EUR 32.240,00
09	Spare parts package for GFM machines	Min. recommended package, for warranty availability	EUR 20.000,00
10	Support for production start	2 experts (el. + mech.) for 1 week (2 men weeks)	EUR 18.570,00

Negotiation allowance: (10%) - 196.500,--

Revised final total price – FOB European Port 1.768.500,--

→ for 1 GFM SKK-10 + 1 pc. Superordinated control (prepared for expansion),
FANUC handling robot + Blank container & spray, Hyfra chiller- big,
Forging tools, pre-acceptance at GFM & assembly at site

These prices are net-fix prices ex works, including dismantling, seaworthy packing, transport insurance, freight charges and all export customs duty, upon FOB European port accord. INCOTERMS 2010. Final destination of installation is Chennai, India.

One-week training at GFM, as well as one-week training at site is included in machine price. Further included are 2 weeks support for production start!

Any additional operator- or maintenance trainingEUR 8.000,00 / week

The indicated prices refer to a design of the machines acc. to GFM-standards. Should there arise any requirements concerning special equipments of the machine after having submitted the offer (especially for hydraulic and electric equipment) which cause higher costs as those of GFM-standard equipment, GFM reserves the right to charge additional costs with indication of justification.

6. CONDITIONS OF SALE

6.1 WARRANTY

GFM guarantees satisfactory function of the machines and equipment for a period of 12 months, commencing after start-up of the machine and equipment at customer's plant but not later than 18 months after shipment resp. readiness for delivery if the start-up is delayed by reasons beyond our responsibility.

The warranty does not cover parts, which are exposed to wear, such as forging tools as well as parts, which are damaged through improper handling.

The warranty does not cover secondary- or consequential losses by any means.

6.2 CONDITIONAL SALE

GFM will be the owner of the equipment until it is completely paid by the customer. If partial payments before delivery are delayed, GFM has the right to delay the delivery date, accordingly.

If payments after delivery will be delayed, GFM has the right to deactivate the machine by software measures.

6.3 PAYMENT

30 % down payment after issue of purchase order, by T/T.

70 % by an irrevocable L/C, established no later than 6 months before shipment in favour of GFM GmbH, thereof:

- 55 % of contract value payable after FOB.
- 15 % of contract value payable on FAC completed, by T/T.

Payment terms: 30 days net

6.4 DELIVERY TIME

11 ½ months after receipt of order and clarification of all technical and commercial details. **FOB European port!**

6.5 VALIDITY OF OFFER

This offer is valid 2 months after date of issue.

The validity of this revised offer #801928/Rev.1 is extended to Dec. 31st, 2021

6.6 ATTACHMENTS

Following documents are part of this offer:

- 1P02296.20.02 Layout for Forging Machine, SKK-10
- 1P02296.16.15a Forging proposals for MTS drive shaft
- 1P02296.16.16a Forging proposals for MTS drive shafts
- Machine- and shaft picture: 2410.04_f01
- General delivery condition: gfm.engl.alb

7. INCLUSIONS

The stated scope of delivery for the machine and equipment includes:

- see pt. 5 for details.

7.1 ASSEMBLY AT OUR PLANT

The machine and equipment will be completely assembled in our plant for required functional test run.

7.2 PRE ACCEPTANCE

A pre-acceptance test run of the machine, including MFU evaluation is optionally provided in our plant in the presence of a representative of the customer.

The no. of work pieces, required for developing, prototype forging and pre-acceptance test run are subject to negotiations.

The blank tubes for those work pieces have to be provided by the customer to our plant free of charge at least one 2 months prior to the date of pre-acceptance.

7.3 DOCUMENTATION

- 7.3.1** A layout drawing of the SKK-10 machine will be delivered by the seller with the Contract in two copies. The buyer will return one approved and signed copy no later than 2 months after contract date.
- 7.3.2** A complete sets of preliminary instructions for installation, assembly, start-up operation, maintenance, repair of machine including description of PC shall be provided by the seller within one month prior to delivery for information only.
- 7.3.3** 2 complete sets of operating manuals as described will be delivered on CD-ROM or USB-Stick, no later than 2 months after final machine runoff.
- a) 1 complete set of technical instructions for installation, assembly, start-up operations maintenance and repair of machine including one set drawings of the machine groups, electric schematic and circuit diagrams, schematic for hydraulic, lubrication and compressed air and relevant technical documentation.
 - b) 1 complete set of detailed drawings for all tooling delivered with the machine including material specifications.
 - c) Description of PC and CNC-control.
 - d) Robot description and manual
 - e) Complete technical descriptions of commercial items
 - f) A complete list of spare parts of the machine including drawings and documents of quick wear parts.
 - g) Oil list with description of viscosity and amount of each media quantity.

8. EXCLUSIONS

1. Bay and cranes as well as floor bound transport devices (fork lift etc.) and transport means (ropes, chains, C-hooks).
2. Illumination
3. Communication system
4. Energy and mediums: supply and discharge (current, cooling water, compressed air) up to the connection points of the GFM supply.
5. Cable trays in the foundation range
6. Fire protection systems
7. Oil fillings, acc. to GFM specification.
8. Mobile cranes and auxiliary means for assembly, unpacking of machine parts and internal transportation (under supervision of GFM-specialists).
9. Workshop equipment for regenerating the forging tools, acc. to GFM advice.

Sincerely yours,

GFM GmbH

A handwritten signature in blue ink is written over a blue ink stamp. The stamp contains the text 'GFM GmbH' and 'A-4403 Steyr, Ennsstrasse 14'.

Franz Hofer, V.P. Sales